2.3 Assignment: Machine Learning vs. Deep Learning

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Choosing the Two Examples

The lesson separates the two approaches by who does the feature work. In traditional machine learning (ML), people choose the features, such as word counts or contract length, and an algorithm like a decision tree or a support vector machine (SVM) learns from them. In deep learning (DL), a neural network with many layers learns its own features from raw data, which takes much more data and computing power (Indiana Wesleyan University, n.d.). To test that difference, I chose one example of each from the lesson: customer churn prediction with an SVM, and machine translation with a deep neural network. For each one, I looked for a published real-world application and asked two questions: why does this approach fit the problem, and why would the other one fit it poorly?

Machine Learning Example: Customer Churn Prediction

Churn prediction means estimating which customers are likely to cancel so that a company can try to keep them. The lesson uses a telecom company that feeds customer age, contract length and monthly charges into an SVM (Indiana Wesleyan University, n.d.). A published application of the same idea comes from Coussement and Van den Poel (2008), who built SVM churn models for a newspaper subscription service. They compared the SVMs with logistic regression and random forests. The SVMs handled the noisy marketing data well, but they beat logistic regression only when their parameters were tuned with the better of two selection methods, and random forests performed best overall. The strongest predictors described the subscription itself, with customer-company interaction variables also playing a role.

Traditional ML suits this problem for three reasons. First, the data is already structured: each customer is a row, and the columns are features that the business understands and can

define, such as subscription length, payment method or complaint history. There is no raw signal, like pixels or sound, that a deep network would need to decode. Second, a churn dataset is usually measured in thousands of customers rather than millions of examples, and the lesson points out that traditional ML can work well with smaller datasets and less processing power. Third, a retention team has to act on the result. A model built on named features makes it easier to explain why a customer was flagged and to retrain the model when conditions change.

DL is a poor fit here. A deep network would need far more labeled churn cases to learn reliably, and with a modest dataset it is more likely to overfit. It would also be harder to explain to managers who want to know which factors drive cancellations. Because the useful information is already captured in a small set of known fields, there is little for extra layers to discover. The study points the same way: the biggest gains came from tuning the model well and from the subscription variables themselves, not from model complexity.

Deep Learning Example: Machine Translation

The lesson lists machine translation as a DL task, where a recurrent neural network or a transformer processes text as a sequence and learns context and meaning (Indiana Wesleyan University, n.d.). A real-world application is Google's Neural Machine Translation system (GNMT), which Wu et al. (2016) describe as a production system at Google. It uses a deep long short-term memory (LSTM) network with eight encoder layers and eight decoder layers, an attention mechanism that links each output word to the relevant parts of the input sentence, and "wordpieces," small sub-word units that help it deal with rare words. In a human side-by-side evaluation on simple sentences, GNMT reduced translation errors by an average of 60% compared with Google's earlier phrase-based production system (Wu et al., 2016).

DL suits translation because the input is raw language, and meaning depends on word order and context. The same word can need a different translation depending on the rest of the sentence, and no team could write a complete list of features covering every phrase in every language pair. A deep network learns those patterns from very large collections of translated sentence pairs, and the attention mechanism lets it decide which source words matter for each word it produces. The output is also a full sentence, not a single label or number.

Traditional ML is not suitable for this task. A model such as logistic regression or an SVM needs a fixed set of features and predicts one class or value. To use it, analysts would have to turn each sentence into hand-made features, like word counts, which throw away the word order that translation depends on. The model still would not be able to generate a new sentence in another language. The size of GNMT's improvement over the earlier system shows how much the learned, layered approach added for this kind of problem, even though it needs the large data and computing resources that the lesson warns about.

Conclusion

Working through these two cases made the difference concrete for me. When the data is structured, the features are already meaningful and the decision has to be explained, traditional ML is the practical choice, as churn prediction shows. When the input is raw and unstructured, the patterns are too complex to list by hand, and large amounts of data are available, DL is worth its cost, as machine translation shows. The right approach depends less on which method is newer and more on the type of data, the amount of data and how the result will be used.

References

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Claude (Anthropic) was used to find and check the two published sources, draft this report from the course lesson and those sources, and check the APA formatting. I reviewed the examples and explanations, edited the wording, and checked the final version against the assignment rubric.